



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Equivalent Expressions

Lesson 6-6 • Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can show that two expressions are equivalent by simplifying and combining like terms.

Language Objective: I can explain my work using the words equivalent, simplify, like terms, and combine.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent

Simplify

Like Terms

Combine

Coefficient



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Expressions that always have the same value.

2

To write an expression in its shortest equal form is to it.

3

Terms with the same variable raised to the same power, like $4x$ and $9x$, are

 .

4

To add or subtract like terms into one term is to them.

5

A number multiplied by a variable is a .

Reflect — Exit Ticket

Which expression is equivalent to $6n + 4 + 3n - 1$?

- A. $9n + 3$
- B. $9n + 5$
- C. $63n$
- D. $18n + 4$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent
2. **Notes 2:** Simplify
3. **Notes 3:** Like Terms
4. **Notes 4:** Combine
5. **Notes 5:** Coefficient
6. **Watch for this mistake:** *A common mistake in Equivalent Expressions is skipping the key idea: "Equivalent expressions give the same answer for every value of the variable." — always check your work against this rule before you submit.*
7. **Try It 1:** A. $4x - 3x + x$ is 3 groups of x plus 1 more group of $x = 4x$. Add the coefficients: $3 + 1 = 4$.
Check $x = 3$: $3(3) + 3 = 12$ and $4(3) = 12$.
8. **Try It 2:** A. $2x + 5$ — Order of addition does not change the value (commutative property): $5 + 2x = 2x + 5$. Check $x = 10$: $5 + 2(10) = 25$ and $2(10) + 5 = 25$.
9. **Exit Ticket:** A. $9n + 3 - 6n + 3n = 9n$ and $4 - 1 = 3$, so the simplified expression is $9n + 3$.